

PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. What is the dimensional formula of $\frac{1}{\mu_0 \epsilon_0}$ (where μ_0 is permeability and ϵ_0 is permittivity of free space).

- (1) LT^{-1} (2) L^2T^{-2}
(3) MLT^{-1} (4) ML^2T^{-2}

Answer (2)

Sol. $\frac{1}{\sqrt{\mu_0 \epsilon_0}} = C$

$$\frac{1}{\mu_0 \epsilon_0} = C^2$$

2. An equilateral prism is made of a material of refractive index $\sqrt{2}$. Find angle of incidence for minimum deviation of the light ray.

- (1) 60° (2) 30°
(3) 37° (4) 45°

Answer (4)

Sol. $\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}}$

$$\frac{60^\circ + \delta_m}{2} = 45^\circ$$

$$\delta_m = 30^\circ$$

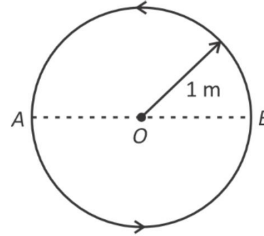
$$\delta_m = i + e - A$$

$$30 = 2i - 60$$

$$i = 45^\circ$$

$$(i = e)$$

3. A particle moves on a circular path of radius 1 m. Find its displacement when it moves from $A \rightarrow B \rightarrow A \rightarrow B$. Also its distance as it moves from $A \rightarrow B \rightarrow A \rightarrow B \rightarrow A$.



- (1) Distance = 2 m, displacement = 4π m
(2) Distance = 2 m, displacement = 5π m
(3) Distance = 4π m, displacement = 2 m
(4) Distance = 5π m, displacement = 2 m

Answer (3)

Sol. Displacement = Shortest distance between final and initial positions = 2 m (One and half cycle)

$$\begin{aligned} \text{Distance} &= \text{Total path length covered} \\ &= 4\pi \text{ m (Two cycles)} \end{aligned}$$

4. The moment of inertia of a ring of mass M and radius R about an axis passing through tangential point in the plane of ring is

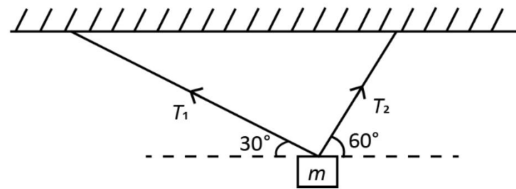
- (1) $\frac{5MR^2}{2}$ (2) $\frac{3MR^2}{2}$
(3) $\frac{4MR^2}{3}$ (4) $\frac{2MR^2}{3}$

Answer (2)

Sol. $I = \frac{MR^2}{2} + MR^2 = \frac{3MR^2}{2}$

5. A block of mass m is suspended in a vertical plane with the help of two light strings as shown. Find the ratio of tensions

$$\frac{T_1}{T_2}$$



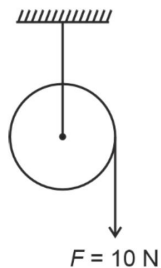
- (1) 3
(2) $\frac{1}{3}$
(3) $\frac{1}{\sqrt{3}}$
(4) $\sqrt{3}$

Answer (3)

Sol. $T_1 \cos 30^\circ = T_2 \cos 60^\circ$

$$\frac{T_1}{T_2} = \frac{\cos 60^\circ}{\cos 30^\circ} = \frac{1}{\sqrt{3}}$$

6. A disc of mass M and radius 2 m is hinged keeping axis horizontal. If angular acceleration of disc is 2 rad/s^2 . Find moment of inertia



- (1) 10 kg m^2
(2) 5 kg m^2
(3) 6 kg m^2
(4) 20 kg m^2

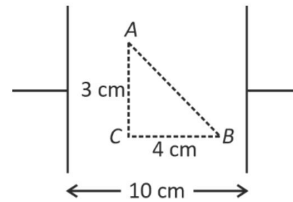
Answer (1)

Sol. $\tau = I\alpha$

$$10 \times 2 = 2I$$

$$I = 10 \text{ kg m}^2$$

7. The figure shows the plates of a parallel plate capacitor with a separation 10 cm and charged to a potential difference V . Find the potential difference between B and A.



- (1) $\frac{2V}{5}$
(2) $\frac{V}{2}$
(3) $\frac{3V}{5}$
(4) $\frac{V}{5}$

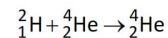
Answer (1)

Sol. $V = E(10 \text{ cm})$

$$V' = E(4 \text{ cm})$$

$$V' = \frac{2}{5}V$$

8. Binding energy per nucleon in ${}^2_1\text{H}$ is x and for ${}^4_2\text{He}$ is y . Find energy released in the given reaction

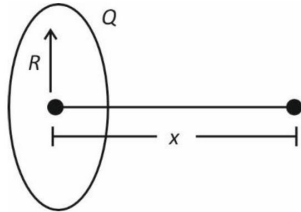


- (1) $2x - 2y$
(2) $-4x + 4y$
(3) $4x - 4y$
(4) $2y - 4x$

Answer (2)

Sol. $BE = 4y - (2x + 2x)$
 $= 4y - 4x$

9. Figure shows a uniformly charged ring having charge Q and radius R . Find the distance from the centre on the axis of the ring where electric field is maximum



- (1) $R\sqrt{2}$ (2) $\frac{R}{\sqrt{2}}$
(3) $2R$ (4) R

Answer (2)

Sol. $E = \frac{kQx}{(R^2 + x^2)^{3/2}}$

$$\frac{dE}{dx} = 0$$

$$x = \frac{R}{\sqrt{2}}$$

10. Two identical drops of radius R and surface tension ' T ' coalesce to form a bigger drop. The change in surface energy in this process is

- (1) $4\pi R^2 T \left[1 - 2^{\frac{1}{3}} \right]$ (2) $8\pi R^2 T \left[1 + 2^{\frac{1}{3}} \right]$
(3) $4\pi R^2 T \left[1 + 2^{\frac{1}{3}} \right]$ (4) $8\pi R^2 T \left[2^{\frac{1}{3}} - 1 \right]$

Answer (4)

Sol. Volume of bigger drop $= \frac{4}{3}\pi R_1^3 = 2 \left(\frac{4}{3}\pi R^3 \right)$

$$R_1 = R(2)^{\frac{1}{3}}$$

Initial energy $= (4\pi R^2 T) \times 2$

Final energy $= 4\pi R^2 (2)^{\frac{2}{3}} T$

11. Two galvanometers G_1 and G_2 are having resistors $R_1 = 5\Omega$ and $R_2 = 7\Omega$, number of turns $N_1 = 21$, $N_2 = 15$, magnetic fields $B_1 = 0.25$ T, $B_2 = 0.50$ T and area of coil $A_1 = 3.6 \times 10^{-3} \text{ cm}^2$ and $A_2 = 1.8 \times 10^{-3} \text{ cm}^2$. Find the ratio of their voltage sensitivity

- (1) $\frac{49}{25}$ (2) $\frac{7}{5}$
(3) $\frac{5}{7}$ (4) $\frac{49}{20}$

Answer (1)

Sol. $\tau = NIAB = K\theta$

$$\frac{\theta}{V} = \frac{\theta}{RI} = \frac{NAB}{LKR}$$

$$\begin{aligned} \text{Ratio of voltage sensitivity} &= \left(\frac{N_1 A_1 B_1}{N_2 A_2 B_2} \right) \frac{R_2}{R_1} \\ &= \frac{21}{15} \times \frac{3.6}{1.8} \times \frac{0.25}{0.50} \times \frac{7}{5} \\ &= \frac{49}{25} \end{aligned}$$

12. Match the List-I with the List-II

- (i) Heat capacity (a) $\text{J kg}^{-1} \text{K}^{-1}$
(ii) Specific heat capacity (b) J K^{-1}
(iii) Latent heat (c) $\text{W m}^{-1} \text{K}^{-1}$
(iv) Thermal conductivity (d) J kg^{-1}
(1) (i)-(b), (ii)-(d), (iii)-(c), (iv)-(a)
(2) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(a)
(3) (i)-(b), (ii)-(c), (iii)-(d), (iv)-(a)
(4) (i)-(b), (ii)-(a), (iii)-(d), (iv)-(c)

Answer (4)

13. In a system of measurement, electric charge (Q), permeability (μ_0) and electric current (i) are considered as fundamental quantity. The dimension of linear momentum in this system is

(1) $[Q^2\mu_0^2i]$ (2) $[Q\mu_0i]$
 (3) $[Q\mu_0i^2]$ (4) $[Q^2\mu_0i]$

Answer (2)

Sol. Let $P \propto (Q)^a (\mu_0)^b (i)^c$

$$[MLT^{-1}] = K [M^b L^b T^{a-2b} A^{a-2b+c}]$$

$$a = 1, b = 1, c = 1$$

14. Which of the following items (labelled i, ii, iii, iv and v) are true

When an ideal gas undergoes adiabatic process, (symbols have their usual meaning)

- (i) $\Delta U = 0$
 (ii) $w = -\Delta U$
 (iii) $PV = \text{Constant}$
 (iv) $VT = \text{Constant}$
 (v) $W \propto |T_2 - T_1|$

- (1) (i), (ii), (iv)
 (2) (ii) and (v)
 (3) (ii), (iii), (v)
 (4) (i), (ii), (v)

Answer (2)

Sol. $\Delta Q = 0$

$$\Rightarrow W = -\Delta U \\ = -nC_V (T_2 - T_1)$$

15. A wave is travelling along a string. The wavelength (λ) of the wave is 7.5 m and amplitude is 2 cm. At $t = 0$, there is a crest at $x = 0$ and in 0.3 seconds it travels a distance of 12 cm in +ve x-direction. The equation of the wave is

(1) $2\sin\left(\frac{2\pi}{15}x + \frac{6\pi}{25}t\right)$ cm (2) $2\cos\left(\frac{4\pi}{15}x - \frac{8\pi}{75}t\right)$ cm
 (3) $2\cos\left(\frac{4\pi}{15}x + \frac{6\pi}{25}t\right)$ cm (4) $2\sin\left(\frac{4\pi}{15}x - \frac{8\pi}{75}t\right)$ cm

Answer (2)

Sol. $\lambda = 7.5$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{7.5} = \frac{4\pi}{15}$$

$$v = \frac{12}{0.3} = 40 \text{ cm/s}$$

$$\frac{\omega}{k} = 40 \text{ cm/s}$$

$$\omega = \frac{40}{100} \times \frac{4\pi}{15} = \frac{40\pi}{375} = \frac{8\pi}{75}$$

$$y = 2\cos\left(\frac{4\pi}{15}x - \frac{8\pi}{75}t\right) \text{ cm}$$

16. An equiconvex lens of radius $R = \frac{1}{6}$ m is having power P .

Another Bi convex lens of radii R_1 and R_2 is having same power P , then

(1) $R_1 = \frac{1}{9}$ m, $R_2 = \frac{1}{3}$ m (2) $R_1 = \frac{1}{6}$ m, $R_2 = \frac{1}{3}$ m
 (3) $R_1 = \frac{1}{9}$ m, $R_2 = \frac{1}{4}$ m (4) $R_1 = \frac{1}{4}$ m, $R_2 = \frac{1}{5}$ m

Answer (1)

$$\text{Sol. } \frac{1}{f_1} = (\mu - 1) \left(\frac{2}{R} \right) = (\mu - 1)12$$

$$\frac{1}{f_2} = (\mu - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = (\mu - 1)12$$

$$\frac{1}{R_1} + \frac{1}{R_2} = 12$$

17. The area of a solenoid is A , length is L , magnetic field inside is B_0 and the relative permeability of medium is 2. The energy stored due to the magnetic field is

$$(1) \frac{B_0^2 AL}{2\mu_0} \quad (2) \frac{B_0^2 AL}{4\mu_0}$$

$$(3) \frac{4B_0^2 AL}{\mu_0} \quad (4) \frac{2B_0^2 AL}{\mu_0}$$

Answer (2)

Sol. $E = \frac{B^2}{2\mu} \times AL$

$$= \frac{B_0^2 AL}{4\mu_0}$$

18.

19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The radius of first Bohr orbit of Li^{2+} is $\frac{a_0}{X}$, where a_0 is the radius of the first Bohr orbit of H . Find X

Answer (3)

Sol. $r = a_0 \frac{n^2}{Z}$

For Li^{2+} is ground state $n = 1$ and $Z = 3$

$$\Rightarrow r = a_0 \frac{(1)^2}{3} = \frac{a_0}{3}$$

22. The length of the string is 104 m when the tension in it is 5 N. The length becomes 1.56 m when the tension in it is 7 N. The natural length of the string is ____ m.

Answer (1)

Sol. $T = k(l - l_0)$

$$\Rightarrow 5 = k(1.4 - l_0)$$

$$\Rightarrow 7 = k(1.56 - l_0)$$

$$\Rightarrow 7(1.4 - l_0) = 5(1.56 - l_0)$$

$$l_0 = \frac{7(1.4) - 5(1.56)}{2} = 1 \text{ m}$$

23. A concave mirror and a convex mirror of same focal length are given. A real object is placed in front of the mirror at a distance equal to half the focal length. The ratio of lateral magnification in the image produced by concave mirror to that produced by the convex mirror is

Answer (3)

Sol. $m = \frac{f}{f - u}$

$$m_{\text{concave}} = \frac{-f}{-f - \left(-\frac{f}{2}\right)} = 2$$

$$m_{\text{convex}} = \frac{+f}{+f - \left(-\frac{f}{2}\right)} = \frac{2}{3}$$

$$\frac{m_{\text{concave}}}{m_{\text{convex}}} = 3$$

24.

25.